

We will cover the following four topics in the Week 2 lectures:

- 1 The Pythagorean distance function.
- 2 Limits.
- 3 Continuity.
- 4 Partial derivatives.

The Pythagorean Distance Function

Let $n \in \mathbb{N}$.

Definition (the Pythagorean distance function)

The **Pythagorean distance function** for $\mathbb{R}^n$ is the function

$$\text{dist} : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$$

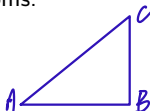
defined by

2 inputs

$$\forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n : \text{dist}(\mathbf{x}, \mathbf{y}) \stackrel{\text{df}}{=} \|\mathbf{x} - \mathbf{y}\|_2.$$

The Pythagorean distance function satisfies the following three axioms:

- ① For all $\mathbf{x}$ in $\mathbb{R}^n$, we have $\text{dist}(\mathbf{x}, \mathbf{x}) = 0$.
- ② For all $\mathbf{x}, \mathbf{y}$ in $\mathbb{R}^n$, we have $\text{dist}(\mathbf{x}, \mathbf{y}) = \text{dist}(\mathbf{y}, \mathbf{x})$.
- ③ For all $\mathbf{x}, \mathbf{y}, \mathbf{z}$ in $\mathbb{R}^n$, we have $\text{dist}(\mathbf{x}, \mathbf{z}) \leq \text{dist}(\mathbf{x}, \mathbf{y}) + \text{dist}(\mathbf{y}, \mathbf{z})$.



Property (2) is known as **symmetry**, and Property (3) the **Triangle Inequality**.

Theorem (challenge time, folks!)

Any function $d : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ that satisfies the three axioms above also satisfies

$$\forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n : d(\mathbf{x}, \mathbf{y}) \geq 0.$$

dist has to be non-negative

Limit Points

Definition (Limit Points)

Let S be a subset of $\mathbb{R}^n$. *n coordinates*

A point a in $\mathbb{R}^n$ is said to be a **limit point** of S in $\mathbb{R}^n$ if and only if the following holds:

For any real number $\epsilon > 0$, there exists an $x \in S$ such that $0 < \text{dist}(x, a) < \epsilon$.
 $x \neq a$
you can find a point in S as close to a as you want

Informally, if S is a subset of $\mathbb{R}^n$, then a point a in $\mathbb{R}^n$ is a limit point of S in $\mathbb{R}^n$ if and only if there are points in S not equal to a that approach a as closely as we want.

Example

- 1 Is $\sqrt{2}$ a limit point of the set of rational numbers in $\mathbb{R}$? *Yes*
- 2 Is $(0, 0)$ a limit point of the unit circle in $\mathbb{R}^2$? *No. Unit circle is always distance 1 away from origin. cannot get as close as we want.*
- 3 Is $(1, 1)$ a limit point of $\{(1, 1), (2, 2)\}$ in $\mathbb{R}^2$?
No. Dist between $(1, 1)$ and $(2, 2)$ is fixed.

Real-Valued Limits — an Informal Description

- Let $m \in \mathbb{N}$.
- Let f be a real-valued function of m real variables. $\rightarrow f$ takes m real inputs and gives a real number value
- Let $S \subseteq \text{Dom}(f)$, and let a be a limit point of S in $\mathbb{R}^m$.
- Let L be a real number.

Then we write

$$\lim_{\substack{x \in S \\ x \rightarrow a}} f(x) = L$$

$\rightarrow x$ should only approach a , not be a

to indicate that as x approaches a within $S \setminus \{a\}$, the value of $f(x)$ approaches L .

If $S = \text{Dom}(f)$, then we simply write $\lim_{x \rightarrow a} f(x) = L$.

Real-Valued Limits — Too Many Directions in Higher Dimensions

A point a on the real line $\mathbb{R}$ can be approached from only two directions: left and right.

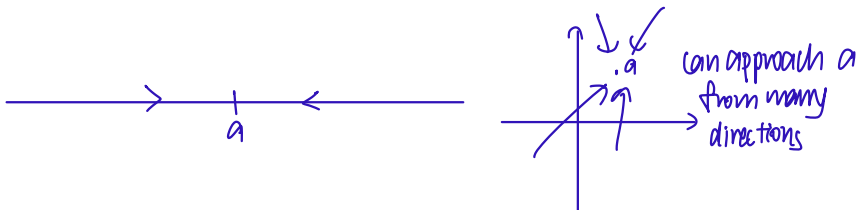
For $m \in \mathbb{N}_{\geq 2}$, a point $\mathbf{a}$ in m -dimensional space $\mathbb{R}^m$, however, can be approached from **infinitely many directions** (and infinitely many styles)!

The limit operation, when given as inputs

- a real-valued function f of several real variables and
- a limit point $\mathbf{a}$ of $\text{Dom}(f)$,

does not care how you approach $\mathbf{a}$ (unless you want it to by a restriction).

It cares only about how closely you approach $\mathbf{a}$, which is quantified by the Pythagorean distance function.



Definition (the ϵ - δ Definition of Real-Valued Limits)

- Let $m \in \mathbb{N}$.
- Let f be a real-valued function of m real variables.
- Let $S \subseteq \text{Dom}(f)$. *Restriction*
- Let $\mathbf{a}$ be a limit point of S in $\mathbb{R}^m$.

Then there exists at most one real number L with the following property:

For each real number $\epsilon > 0$, there exists a real number $\delta > 0$ such that

$$\forall \mathbf{x} \in S : 0 < \text{dist}(\mathbf{x}, \mathbf{a}) < \delta \implies |f(\mathbf{x}) - L| < \epsilon.$$

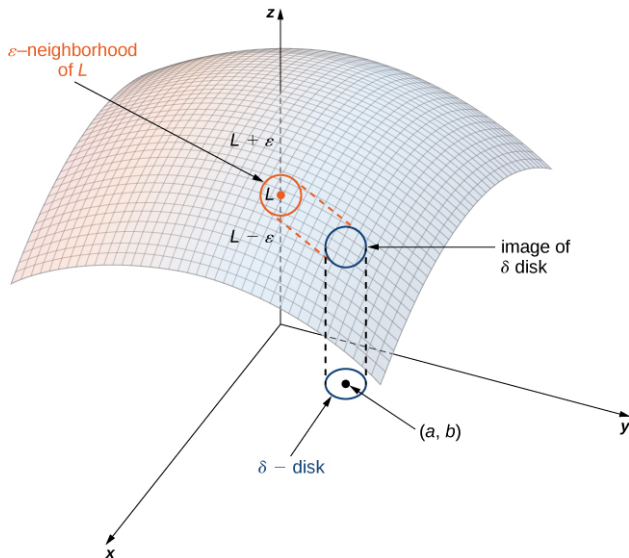
Handwritten notes: $\text{dist}(\mathbf{x}, \mathbf{a})$ is circled with a note " x is not a !". An arrow points from $|f(\mathbf{x}) - L|$ to the text "Pythagorean distance".

If such a real number L indeed exists, then we shall denote it by $\lim_{\substack{\mathbf{x} \in S \\ \mathbf{x} \rightarrow \mathbf{a}}} f(\mathbf{x})$ and call it the S -restricted limit of f at $\mathbf{a}$.

Handwritten notes: A box around the limit notation has a note " x is in S ".

If $S = \text{Dom}(f)$ and $\lim_{\substack{\mathbf{x} \in S \\ \mathbf{x} \rightarrow \mathbf{a}}} f(\mathbf{x})$ exists, then we simply write it as $\lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x})$ and call it the limit of f at $\mathbf{a}$.

Real-Valued Limits — a Visualization



In the ϵ - δ definition of real-valued limits, note the following:

- $f(\mathbf{a})$ may not be defined because $\mathbf{a}$ may not be in $\text{Dom}(f)$ (it must, however, be a limit point of $\text{Dom}(f)$ in $\mathbb{R}^m$).
- $|f(\mathbf{x}) - L|$ is actually the Pythagorean distance between $f(\mathbf{x})$ and L in $\mathbb{R}$:

$$\forall x, y \in \mathbb{R} : \quad \text{dist}(x, y) = \sqrt{(x - y)^2} = |x - y|.$$

Real-Valued Limits — the Two Categories of Problems

Problems on limits fall into the following two categories:

- Show that the limit of f at a exists.
- Show that the limit of f at a does not exist.

* typically it is easier to
show that limit does
not exist

We will employ various strategies to solve problems in each category.

Theorem (a Non-Existence Result)

- Let f be a real-valued function of two real variables.
- Let $P, Q \subseteq \text{Dom}(f)$ (usually intersections of curves in $\mathbb{R}^2$ with $\text{Dom}(f)$).
- Let a be a limit point of both P and Q in $\mathbb{R}^2$. *a must be a limit point*
- Suppose that one of the following statements holds:
 - Either $\lim_{\substack{x \in P \\ x \rightarrow a}} f(x)$ or $\lim_{\substack{x \in Q \\ x \rightarrow a}} f(x)$ does not exist.
 - Both $\lim_{\substack{x \in P \\ x \rightarrow a}} f(x)$ and $\lim_{\substack{x \in Q \\ x \rightarrow a}} f(x)$ exist but are different.

} restricted limits, only within P & Q

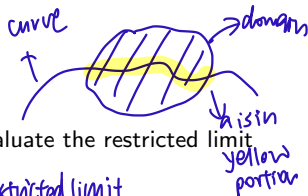
Then $\lim_{x \rightarrow a} f(x)$ does not exist. *← unrestricted*

Caution: This theorem is a non-existence result only and cannot be used to show that real-valued limits in two dimensions exist!

Real-Valued Limits in Two Dimensions — Showing Non-Existence

If you suspect that a real-valued limit in two dimensions does not exist, then attempt to show that the restricted limits on two different curves (it takes only two) are different.

- Let f be a real-valued function of two real variables.
- Let C be a curve.
- Let $\mathbf{a} = (a, b)$ be a limit point of $C \cap \text{Dom}(f)$ in $\mathbb{R}^2$.



The table below shows, for various forms of C , how to evaluate the restricted limit

$$\lim_{\substack{\mathbf{x} \in C \cap \text{Dom}(f) \\ \mathbf{x} \rightarrow \mathbf{a}}} f(\mathbf{x}) \rightarrow \text{restricted limit}$$

as a real-valued limit in one dimension.

Equation of C	$\lim_{\substack{\mathbf{x} \in C \cap \text{Dom}(f) \\ \mathbf{x} \rightarrow \mathbf{a}}} f(\mathbf{x})$ becomes
$x = a$	$\lim_{y \rightarrow b} f(a, y)$
$y = b$	$\lim_{x \rightarrow a} f(x, b)$
$x = g(y)$	$\lim_{y \rightarrow b} f(g(y), y)$
$y = g(x)$	$\lim_{x \rightarrow a} f(x, g(x))$

Calc I Problem

→ everything depends on y here
→ everything depends on x here

Example

Let $D \stackrel{\text{df}}{=} \{(x, y) \in \mathbb{R}^2 \mid x + y \neq 1\}$, and define $f : D \rightarrow \mathbb{R}$ by $f(x, y) \stackrel{\text{df}}{=} \frac{y}{x + y - 1}$.

Determine whether $\lim_{(x, y) \rightarrow (1, 0)} f(x, y)$ exists.

Solution

- Restrict the limit to the line given by the equation $x = 1$:

$$\lim_{y \rightarrow 0} \frac{y}{1 + y - 1} = \lim_{y \rightarrow 0} \frac{y}{y} = \lim_{y \rightarrow 0} 1 = 1.$$

- Restrict the limit to the line given by the equation $y = 0$:

$$\lim_{x \rightarrow 1} \frac{0}{x + 0 - 1} = \lim_{x \rightarrow 1} \frac{0}{x - 1} = \lim_{x \rightarrow 1} 0 = 0.$$

- As these two restricted limits are different, the limit itself does not exist.

$\downarrow$
 $\neq 0$

Example

Define $f : \mathbb{R}^2 \setminus \{(0, 0)\} \rightarrow \mathbb{R}$ by $f(x, y) \stackrel{\text{df}}{=} \frac{xy}{x^2 + y^2}$.

Determine whether $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ exists.

Solution

- Restrict the limit to the line given by the equation $x = 0$:

$$\lim_{y \rightarrow 0} \frac{0 \cdot y}{0^2 + y^2} = \lim_{y \rightarrow 0} \frac{0}{y^2} = \lim_{y \rightarrow 0} 0 = 0.$$

- Restrict the limit to the line given by the equation $y = 0$:

$$\lim_{x \rightarrow 0} \frac{x \cdot 0}{x^2 + 0^2} = \lim_{x \rightarrow 0} \frac{0}{x^2} = \lim_{x \rightarrow 0} 0 = 0.$$

- These two restricted limits are equal, but it DOES NOT mean that the limit itself exists and is equal to 0!
- There are infinitely many other paths to consider.

Real-Valued Limits in Two Dimensions — Example 2 (cont'd)

- Restrict the limit to the line given by the equation $y = x$:

$$\lim_{x \rightarrow 0} \frac{x^2}{x^2 + x^2} = \lim_{x \rightarrow 0} \frac{x^2}{2x^2} = \lim_{x \rightarrow 0} \frac{1}{2} = \frac{1}{2}.$$

- As this restricted limit is different from the previous two, the limit itself does not exist.

Example

Define $f : \mathbb{R}^2 \setminus \{(0, 0)\} \rightarrow \mathbb{R}$ by $f(x, y) \stackrel{\text{df}}{=} \frac{xy^2}{x^2 + y^4}$.

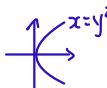
Determine whether $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ exists.

Solution

- Restriction of the limit to either the x -axis or the y -axis yields 0.
- Let $m \in \mathbb{R}$ and restrict the limit to the line given by the equation $y = mx$:

$$\lim_{x \rightarrow 0} \frac{x(mx)^2}{x^2 + (mx)^4} = \lim_{x \rightarrow 0} \frac{m^2 x^3}{x^2 + m^4 x^4} = \lim_{x \rightarrow 0} \frac{m^2 x}{1 + m^4 x^2} = \frac{0}{1 + 0} = 0.$$

- 0 yet again! Yikes! Let us try a non-linear curve this time.

- Restrict the limit to the curve given by the equation $x = y^2$: 

$$\lim_{y \rightarrow 0} \frac{y^2 y^2}{(y^2)^2 + y^4} = \lim_{y \rightarrow 0} \frac{y^4}{y^4 + y^4} = \lim_{y \rightarrow 0} \frac{y^4}{2y^4} = \lim_{y \rightarrow 0} \frac{1}{2} = \frac{1}{2}.$$

- As this restricted limit is different from the previous ones, the limit itself does not exist.

Real-Valued Limits — Showing Existence

We now want to study techniques for establishing that real-valued limits exist.

Showing that a real-valued limit exists when restricted to a few curves is not enough.

There are infinitely many other curves or non-curves that could screw things up.

The unrestricted limit **DOES NOT and CANNOT** care what restrictions are used.

To show that a real-valued limit exists, we can do one of the following:

- Apply well-known limit theorems or the continuity of well-known functions.
- Apply the Squeeze Theorem (in two dimensions, this is sometimes done in concert with the polar-coordinate transformation).

continuous functions: exponential functions, sine & cosine functions

Theorem (Some Limit Results)

- Let $m \in \mathbb{N}$.
- Let f and g be real-valued functions of m real variables with domain D .
- Let $\mathbf{a}$ be a limit point of D in $\mathbb{R}^m$.
- Let $c \in \mathbb{R}$.
- Suppose that both $\lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x})$ and $\lim_{\mathbf{x} \rightarrow \mathbf{a}} g(\mathbf{x})$ exist.

Then the following statements are true:

- $\lim_{\mathbf{x} \rightarrow \mathbf{a}} (f \pm g)(\mathbf{x})$ exists and $\lim_{\mathbf{x} \rightarrow \mathbf{a}} (f \pm g)(\mathbf{x}) = \lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x}) \pm \lim_{\mathbf{x} \rightarrow \mathbf{a}} g(\mathbf{x})$.
- $\lim_{\mathbf{x} \rightarrow \mathbf{a}} (c \cdot f)(\mathbf{x})$ exists and $\lim_{\mathbf{x} \rightarrow \mathbf{a}} (c \cdot f)(\mathbf{x}) = c \cdot \lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x})$.
- $\lim_{\mathbf{x} \rightarrow \mathbf{a}} (f \cdot g)(\mathbf{x})$ exists and $\lim_{\mathbf{x} \rightarrow \mathbf{a}} (f \cdot g)(\mathbf{x}) = \lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x}) \cdot \lim_{\mathbf{x} \rightarrow \mathbf{a}} g(\mathbf{x})$.
- If $\lim_{\mathbf{x} \rightarrow \mathbf{a}} g(\mathbf{x}) \neq 0$, then $\lim_{\mathbf{x} \rightarrow \mathbf{a}} \left(\frac{f}{g} \right)(\mathbf{x})$ exists and $\lim_{\mathbf{x} \rightarrow \mathbf{a}} \left(\frac{f}{g} \right)(\mathbf{x}) = \frac{\lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x})}{\lim_{\mathbf{x} \rightarrow \mathbf{a}} g(\mathbf{x})}$.

Theorem (the Squeeze Theorem)

- Let $m \in \mathbb{N}$.
- Let f , g , and h be real-valued functions of m real variables with domain D .
- Let $\mathbf{a}$ be a limit point of D in $\mathbb{R}^m$.
- Suppose that for all $\mathbf{x} \in D \setminus \{\mathbf{a}\}$ within a sufficiently small distance of $\mathbf{a}$, we have

$$f(\mathbf{x}) \leq g(\mathbf{x}) \leq h(\mathbf{x}).$$

- Suppose that $\lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x})$ and $\lim_{\mathbf{x} \rightarrow \mathbf{a}} h(\mathbf{x})$ exist and that $\lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x}) = L = \lim_{\mathbf{x} \rightarrow \mathbf{a}} h(\mathbf{x})$.

Then $\lim_{\mathbf{x} \rightarrow \mathbf{a}} g(\mathbf{x})$ exists and $\lim_{\mathbf{x} \rightarrow \mathbf{a}} g(\mathbf{x}) = L$.

Example

Define $f : \mathbb{R}^2 \setminus \{(0, 0)\} \rightarrow \mathbb{R}$ by $f(x, y) \stackrel{\text{df}}{=} \frac{\sin(\pi x) + \cos(\pi y)}{x^2 + y^2}$.

Evaluate $\lim_{(x,y) \rightarrow (1,1)} f(x, y)$.

Solution

- Observe that

$$\begin{aligned}\lim_{(x,y) \rightarrow (1,1)} (\sin(\pi x) + \cos(\pi y)) &= \lim_{x \rightarrow 1} \sin(\pi x) + \lim_{y \rightarrow 1} \cos(\pi y) \\ &= \sin(\pi) + \cos(\pi) \\ &= 0 - 1 \\ &= -1.\end{aligned}$$

- Observe that $\lim_{(x,y) \rightarrow (1,1)} (x^2 + y^2) = \lim_{x \rightarrow 1} x^2 + \lim_{y \rightarrow 1} y^2 = 1 + 1 = 2$.
- Therefore, $\lim_{(x,y) \rightarrow (1,1)} f(x, y) = -\frac{1}{2}$.

Example

Define $f : \mathbb{R}^2 \setminus \{(0,0)\} \rightarrow \mathbb{R}$ by $f(x,y) \stackrel{\text{df}}{=} \frac{3x^2y}{x^2+y^2}$. Show that $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0$.

Solution

- For all x and y , if $(x,y) \in \mathbb{R}^2 \setminus \{(0,0)\}$, then

$$0 \leq \left| \frac{3x^2y}{x^2+y^2} \right| = 3 \cdot \frac{x^2}{x^2+y^2} \cdot \underbrace{|y|}_{\text{need absolute sign}} \leq 3 \cdot 1 \cdot |y| = 3|y|. \quad \frac{x^2}{x^2+y^2} \leq 1$$

- Observe that $\lim_{(x,y) \rightarrow (0,0)} 0 = 0$ and $\lim_{(x,y) \rightarrow (0,0)} |y| = \lim_{y \rightarrow 0} |y| = 0$.

- The Squeeze Theorem then yields $\lim_{(x,y) \rightarrow (0,0)} \left| \frac{3x^2y}{x^2+y^2} \right| = 0$.

- Therefore, $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0$.

$$-|x| \leq x \leq |x|$$

$\therefore$ numerator is always less than denominator

Definition (the ϵ - δ Definition of Vector-Valued Limits)

- Let $m, n \in \mathbb{N}$. *each output is a vector*
- Let f be an $\mathbb{R}^n$ -valued function of m real variables.
- Let $S \subseteq \text{Dom}(f)$.
- Let $\mathbf{a}$ be a limit point of S in $\mathbb{R}^m$.

Then there exists at most one vector $\mathbf{L}$ in $\mathbb{R}^n$ with the following property:

For each real number $\epsilon > 0$, there exists a real number $\delta > 0$ such that

$$\forall \mathbf{x} \in S : 0 < \text{dist}(\mathbf{x}, \mathbf{a}) < \delta \implies \text{dist}(f(\mathbf{x}), \mathbf{L}) < \epsilon.$$

→ $\mathbf{x} \neq \mathbf{a}$!

If such a vector $\mathbf{L}$ in $\mathbb{R}^n$ indeed exists, then we shall denote it by $\lim_{\substack{\mathbf{x} \in S \\ \mathbf{x} \rightarrow \mathbf{a}}} f(\mathbf{x})$.

If $S = \text{Dom}(f)$ and $\lim_{\substack{\mathbf{x} \in S \\ \mathbf{x} \rightarrow \mathbf{a}}} f(\mathbf{x})$ exists, then we simply write it as $\lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x})$.

Theorem (an Alternative Characterization of Vector-Valued Limits)

- Let $m, n \in \mathbb{N}$.
- Let $f = (f_1, \dots, f_n)$ be an $\mathbb{R}^n$ -valued function of m real variables.
- Let $\mathbf{a}$ be a limit point of $\text{Dom}(f)$ in $\mathbb{R}^m$.
- Let $\mathbf{L} = (L_1, \dots, L_n)$ be a vector in $\mathbb{R}^n$.

Then the following two statements are equivalent:

- $\lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x}) = \mathbf{L}$. *→ comes from definition in previous slide*
- For each $i \in \{1, \dots, n\}$, we have $\lim_{\mathbf{x} \rightarrow \mathbf{a}} f_i(\mathbf{x}) = L_i$.

$$(f_1(\mathbf{x}), \dots, f_n(\mathbf{x})) \rightarrow (L_1, \dots, L_n)$$

Example

Define $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}$ by $f(x) \stackrel{\text{df}}{=} \left(x^x, \frac{\cos(x) - 1}{x^2} \right)$. Evaluate $\lim_{x \rightarrow 0^+} f(x)$.

Solution

- By l'Hôpital's Rule,

differentiate, L'Hôpital's Rule

$$\lim_{x \rightarrow 0^+} x \ln(x) = \lim_{x \rightarrow 0^+} \frac{\ln(x) \xrightarrow{-\infty}}{\left(\frac{1}{x}\right) \xrightarrow{+\infty}} = \lim_{x \rightarrow 0^+} \frac{\left(\frac{1}{x}\right)}{\left(-\frac{1}{x^2}\right)} = \lim_{x \rightarrow 0^+} (-x) = -0 = 0.$$

- Hence, $\lim_{x \rightarrow 0^+} x^x = \lim_{x \rightarrow 0^+} e^{x \ln(x)} = e^{\lim_{x \rightarrow 0^+} x \ln(x)} = e^0 = 1$.

- By l'Hôpital's Rule again,

differentiate

By continuity

differentiate

$$\lim_{x \rightarrow 0^+} \frac{\cos(x) - 1}{x^2} = \lim_{x \rightarrow 0^+} \frac{-\sin(x)}{2x} = \lim_{x \rightarrow 0^+} \frac{-\cos(x)}{2} = \frac{-\cos(0)}{2} = -\frac{1}{2}.$$

- Therefore, $\lim_{x \rightarrow 0^+} f(x) = \left(0, -\frac{1}{2} \right)$.

Definition (the ϵ - δ Definition of Real-Valued Continuity)

- Let $m \in \mathbb{N}$.
- Let f be a real-valued function of m real variables.
- Let $\mathbf{a} \in \mathbb{R}^m$.

Then f is said to be **continuous** at $\mathbf{a}$ if and only if $\mathbf{a} \in \text{Dom}(f)$ and the following holds:

For each real number $\epsilon > 0$, there exists a real number $\delta > 0$ such that

x can be a here

$$\forall \mathbf{x} \in \text{Dom}(f) : \text{dist}(\mathbf{x}, \mathbf{a}) < \delta \implies |f(\mathbf{x}) - f(\mathbf{a})| < \epsilon.$$

Theorem (an Alternative Characterization of Real-Valued Continuity)

- Let $m \in \mathbb{N}$.
- Let f be a real-valued function of m real variables.
- Let $\mathbf{a} \in \mathbb{R}^m$.

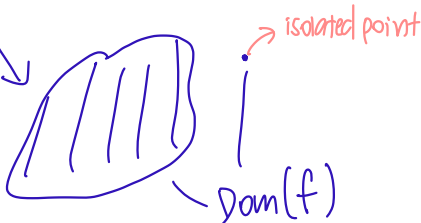
Then f is continuous at $\mathbf{a}$ if and only if $\mathbf{a} \in \text{Dom}(f)$ and one of the following holds:

- $\mathbf{a}$ is not a limit point of $\text{Dom}(f)$ in $\mathbb{R}^m$ (nothing else to check).

Remark: Continuity is therefore automatic at isolated points of $\text{Dom}(f)$ in $\mathbb{R}^m$.

- $\mathbf{a}$ is a limit point of $\text{Dom}(f)$ in $\mathbb{R}^m$ and $\lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x}) = f(\mathbf{a})$.

Remark: This statement is known as the **Substitution Criterion**.



Theorem (Some Continuity Results)

- Let $m \in \mathbb{N}$.
- Let f and g be real-valued functions of m real variables with domain D .
- Let $\mathbf{a} \in \mathbb{R}^m$.
- Let $c \in \mathbb{R}$.
- Suppose that f and g are continuous at $\mathbf{a}$. *Assume!*

Then the following statements are true:

- $f \pm g$ is continuous at $\mathbf{a}$.
- $c \cdot f$ is continuous at $\mathbf{a}$.
- $f \cdot g$ is continuous at $\mathbf{a}$.
- If $g(\mathbf{a}) \neq 0$, then $\frac{f}{g}$ is continuous at $\mathbf{a}$.

Some Classes of Real-Valued Continuous Functions

The following real-valued functions are continuous throughout their domain:

- Multivariate polynomial functions. e.g. $f(x, y) = 3x^2y - x^3y^4$
↳ linear combi of powers of x multiplied by powers of $y \Rightarrow$ natural no. powers
- Multivariate rational functions.
↳ polynomial function divided by polynomial function

Given multivariate polynomial functions p and q of m real variables, the rational function $\frac{p}{q}$ is continuous throughout its domain, given by $q \neq 0$

$$\text{Dom}\left(\frac{p}{q}\right) = \{\mathbf{x} \in \mathbb{R}^m \mid q(\mathbf{x}) \neq 0\}.$$

- Trigonometric or exponential functions composed with the above.
 $\sin, \cos, \tan$ etc

Example

Define $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f(x, y) \stackrel{\text{df}}{=} \begin{cases} \overset{p(x,y)}{x^4}, & \text{if } (x, y) \neq (0, 0); \\ \underset{q(x,y)}{0}, & \text{if } (x, y) = (0, 0). \end{cases}$$

Problem: Find the points of continuity of f .

Solution

- The restriction of f to $\mathbb{R}^2 \setminus \{(0, 0)\}$ is the rational function $r = \frac{p}{q}$, where p and q are the bivariate polynomial functions defined by

$$p(x, y) \stackrel{\text{df}}{=} x^4 \quad \text{and} \quad q(x, y) \stackrel{\text{df}}{=} x^2 + y^4.$$

- As r is continuous on $\mathbb{R}^2 \setminus \{(0, 0)\}$, which is an open subset of $\mathbb{R}^2$, it follows that f is also continuous on $\mathbb{R}^2 \setminus \{(0, 0)\}$.

Real-Valued Continuity — Example 1 (cont'd)

- It only remains to check the continuity of f at $(0, 0)$.
- As $(0, 0)$ is a limit point of $\text{Dom}(f)$, it suffices to check whether

Only can approach this point!

$$\lim_{(x,y) \rightarrow (0,0)} f(x) = f(0, 0) = 0.$$

→ see slide 26

- For all x, y , if $(x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\}$, then

$$0 \leq f(x, y) = \frac{x^4}{x^2 + y^2} = \frac{(x^2)^2}{x^2 + y^2} \leq \frac{(x^2 + y^2)^2}{x^2 + y^2} = x^2 + y^2.$$

- As $\lim_{(x,y) \rightarrow (0,0)} (x^2 + y^2) = 0$, the Squeeze Theorem yields $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0$.
- Therefore, f is continuous on $\mathbb{R}^2$.

Real-Valued Continuity — a Composition Result

Theorem (Composition Result)

- Let f be a real-valued function of m real variables.
- Let g be a real-valued function of one real variable such that $\text{Range}(f) \subseteq \text{Dom}(g)$.
- Let $\mathbf{a}$ be a limit point of $\text{Dom}(f)$ in $\mathbb{R}^m$.
- Let $L \in \mathbb{R}$.
- Suppose that $\lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x}) = L$ and that g is continuous at L .

$$\text{Dom}(g \circ f) = \text{Dom}(f)$$

Then $\mathbf{a}$ is a limit point of $\text{Dom}(g \circ f)$ in $\mathbb{R}^m$ and

$$\lim_{\mathbf{x} \rightarrow \mathbf{a}} (g \circ f)(\mathbf{x}) = g\left(\lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x})\right) = g(L).$$

if g is continuous at the value of the limit for f , we can move the limit inside

Therefore, if f is continuous at $\mathbf{a}$ and g is continuous at $f(\mathbf{a})$, then $g \circ f$ is continuous at $\mathbf{a}$.

Example

Define $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ by $f(x, y, z) \stackrel{\text{df}}{=} e^{x^2 y z^3}$.

Problem: Find the points of continuity of f .

exponential functions are continuous!

Solution

- Define a trivariate polynomial function p by $p(x, y, z) \stackrel{\text{df}}{=} x^2 y z^3$.
- Observe that f is the composition of the exponential function $\exp$ with p .
- As both $\exp$ and p are continuous throughout their domain, f is continuous on $\mathbb{R}^3$.

$\therefore f$ is continuous throughout its domain

if we change exponential to sine, the ans is the same

Vector-Valued Continuity — a Formal Definition

Definition (ϵ - δ Definition of Vector-Valued Continuity)

- Let f be an $\mathbb{R}^n$ -valued function of m real variables.
- Let $\mathbf{a} \in \mathbb{R}^m$.

Then f is said to be **continuous** at $\mathbf{a}$ if and only if $\mathbf{a} \in \text{Dom}(f)$ and the following holds:

For each real number $\epsilon > 0$, there exists a real number $\delta > 0$ such that

$$\forall \mathbf{x} \in \text{Dom}(f) : \text{dist}(\mathbf{x}, \mathbf{a}) < \delta \implies \text{dist}(f(\mathbf{x}), f(\mathbf{a})) < \epsilon.$$

Handwritten notes:
→ $\mathbf{a}$ can be $\mathbf{a}$ here $\implies$ allowed for continuity,
not for limits

Theorem (Two Alternative Characterizations of Vector-Valued Continuity)

- Let $f = (f_1, \dots, f_n)$ be an $\mathbb{R}^n$ -valued function of m real variables.
- Let $\mathbf{a} \in \mathbb{R}^m$.
- Let $\mathbf{L} = (L_1, \dots, L_n) \in \mathbb{R}^n$.

Then f is continuous at $\mathbf{a}$ if and only if $f_1, \dots, f_n$ are continuous at $\mathbf{a}$.

Also, f is continuous at $\mathbf{a}$ if and only if one of the following statements holds:

- $\mathbf{a}$ is an isolated point of $\text{Dom}(f)$ in $\mathbb{R}^m$ (nothing else to check).
- $\mathbf{a}$ is a limit point of $\text{Dom}(f)$ in $\mathbb{R}^m$ and $\lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x}) = f(\mathbf{a})$.

Substitution criterion

①

②

Theorem (Some Continuity Results)

- Let f and g be $\mathbb{R}^n$ -valued functions of m real variables with domain D .
- Let $\mathbf{a}$ be a limit point of D in $\mathbb{R}^m$.
- Let $c \in \mathbb{R}$.
- Suppose that both $\lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x})$ and $\lim_{\mathbf{x} \rightarrow \mathbf{a}} g(\mathbf{x})$ exist. Assume!

Then the following statements are true:

- $\lim_{\mathbf{x} \rightarrow \mathbf{a}} (f \pm g)(\mathbf{x})$ exists and $\lim_{\mathbf{x} \rightarrow \mathbf{a}} (f \pm g)(\mathbf{x}) = \lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x}) \pm \lim_{\mathbf{x} \rightarrow \mathbf{a}} g(\mathbf{x})$. *
- $\lim_{\mathbf{x} \rightarrow \mathbf{a}} (c \cdot f)(\mathbf{x})$ exists and $\lim_{\mathbf{x} \rightarrow \mathbf{a}} (c \cdot f)(\mathbf{x}) = c \cdot \lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x})$.
- $\lim_{\mathbf{x} \rightarrow \mathbf{a}} (f \bullet g)(\mathbf{x})$ exists and $\lim_{\mathbf{x} \rightarrow \mathbf{a}} (f \bullet g)(\mathbf{x}) = \lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x}) \bullet \lim_{\mathbf{x} \rightarrow \mathbf{a}} g(\mathbf{x})$. limits also respect dot product

In particular, if both f and g are continuous at $\mathbf{a}$, then so are $f \pm g$, $c \cdot f$, and $f \bullet g$.

dot product takes 2 vectors and gives a scalar

Vector-Valued Continuity — a Composition Result

Theorem (Composition Result)

- Let f be an $\mathbb{R}^m$ -valued function of l real variables.
- Let g be an $\mathbb{R}^n$ -valued function of m real variables such that $\text{Range}(f) \subseteq \text{Dom}(g)$.
- Let $\mathbf{a}$ be a limit point of $\text{Dom}(f)$ in $\mathbb{R}^l$.
- Let $\mathbf{L} \in \mathbb{R}^m$.
- Suppose that $\lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x}) = \mathbf{L}$ and that g is continuous at $\mathbf{L}$.

$g \circ f$

Then $\mathbf{a}$ is a limit point of $\text{Dom}(g \circ f)$ in $\mathbb{R}^l$ and

$$\lim_{\mathbf{x} \rightarrow \mathbf{a}} (g \circ f)(\mathbf{x}) = g\left(\lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x})\right) = g(\mathbf{L}).$$

Therefore, if f is continuous at $\mathbf{a}$ and g is continuous at $f(\mathbf{a})$, then $g \circ f$ is continuous at $\mathbf{a}$.

Vector-Valued Limits at Infinity — a Formal Definition

Real Valued

Definition (ϵ - N Definition of Vector-Valued Limits at ∞)

- Let f be an $\mathbb{R}^n$ -valued function of m real variables.
- Let $S \subseteq \text{Dom}(f)$.
- Suppose that S is **unbounded**, i.e., regardless of what positive real number r you choose, there is always an $x \in S$ whose distance from $\mathbf{0}_m$ is larger than r .

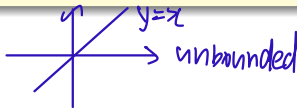
Then there exists at most one vector $\mathbf{L}$ in $\mathbb{R}^n$ with the following property:

For each real number $\epsilon > 0$, there exists a real number $N > 0$ such that

$$\forall \mathbf{x} \in S : \quad \text{dist}(\mathbf{x}, \mathbf{0}_m) > N \implies \text{dist}(f(\mathbf{x}), \mathbf{L}) < \epsilon.$$

If such a vector $\mathbf{L}$ in $\mathbb{R}^n$ indeed exists, then we shall denote it by $\lim_{\substack{\mathbf{x} \in S \\ \mathbf{x} \rightarrow \infty}} f(\mathbf{x})$.

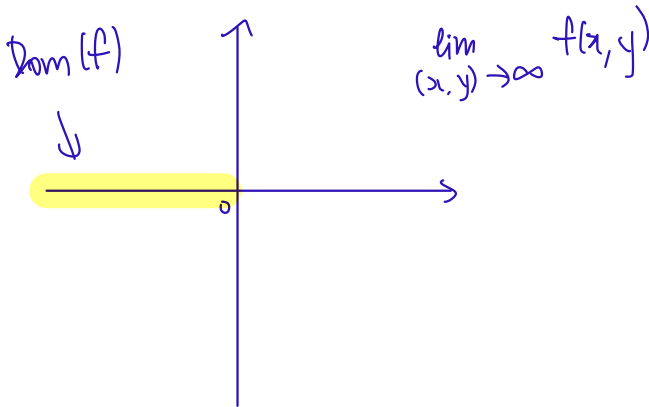
If $S = \text{Dom}(f)$ and $\lim_{\substack{\mathbf{x} \in S \\ \mathbf{x} \rightarrow \infty}} f(\mathbf{x})$ exists, then we simply write it as $\lim_{\mathbf{x} \rightarrow \infty} f(\mathbf{x})$.



Vector-Valued Limits at Infinity — Important Pointers

The ∞ appearing in the ϵ - N definition of vector-valued limits at infinity **DOES NOT** represent the positive direction on either the x -axis or the y -axis.

It means to get as far away from the origin as you possibly can — in any direction.



Example

Define $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $f(x, y) \stackrel{\text{df}}{=} e^{-(x+y)} = e^{-x}e^{-y}$

- (a) Show that $\lim_{x \rightarrow \infty} \left(\lim_{y \rightarrow \infty} f(x, y) \right) = 0$. *Iterated limit $\Rightarrow$ do one after another*
- (b) Show that $\lim_{(x, y) \rightarrow \infty} f(x, y)$ does not exist.

Solution

(a) We have

$$\begin{aligned} \lim_{x \rightarrow \infty} \left(\lim_{y \rightarrow \infty} f(x, y) \right) &= \lim_{x \rightarrow \infty} \left(\lim_{y \rightarrow \infty} e^{-x} e^{-y} \right) \quad \text{Fixed first} \\ &= \lim_{x \rightarrow \infty} \left(e^{-x} \lim_{y \rightarrow \infty} e^{-y} \right) \\ &= \lim_{x \rightarrow \infty} e^{-x} \cdot 0 \quad \begin{array}{c} \parallel \\ 0 \end{array} \\ &= \lim_{x \rightarrow \infty} 0 \\ &= 0. \end{aligned}$$

Vector-Valued Limits at Infinity — Example 1 (cont'd)

(b)

- Restrict the limit to the ~~positive x axis~~ ^{to the ray given by} $\mathbb{R}_{>0} \times \{0\}$:

$$\lim_{x \rightarrow \infty} f(x, 0) = \lim_{x \rightarrow \infty} e^{-x+0} = \lim_{x \rightarrow \infty} e^{-x} = 0.$$

- Restrict the limit to the ~~negative x axis~~ ^{ray given by} $\mathbb{R}_{<0} \times \{0\}$:

$$\lim_{x \rightarrow -\infty} f(x, 0) = \lim_{x \rightarrow -\infty} e^{-x+0} = \lim_{x \rightarrow -\infty} e^{-x} = \infty.$$

- As these two restricted limits are different, the limit itself does not exist.

Example

Define $f : \mathbb{R}^2 \setminus \{(0, 0)\} \rightarrow \mathbb{R}^2$ by $f(x, y) \stackrel{\text{df}}{=} \left(\frac{x}{x^2 + y^2}, -1 \right)$.

Problem: Evaluate $\lim_{(x, y) \rightarrow \infty} f(x, y)$.

Solution

- For all x, y , if $(x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\}$, then

$$0 \leq \left| \frac{x}{x^2 + y^2} \right| = \frac{|x|}{x^2 + y^2} = \frac{\sqrt{x^2}}{x^2 + y^2} \leq \frac{\sqrt{x^2 + y^2}}{x^2 + y^2} = \frac{1}{\sqrt{x^2 + y^2}} = \frac{1}{\text{dist}((x, y), (0, 0))} \rightarrow 0.$$

similar manipulation to slide 30

- By the Squeeze Theorem, $\lim_{(x, y) \rightarrow \infty} \left| \frac{x}{x^2 + y^2} \right| = 0$.

- Therefore, $\lim_{(x, y) \rightarrow \infty} \frac{x}{x^2 + y^2} = 0$.

- Finally, as $\lim_{(x, y) \rightarrow \infty} (-1) = -1$, we conclude that $\lim_{(x, y) \rightarrow \infty} f(x, y) = (0, -1)$.

dist function from origin

dist gets bigger and bigger

Open and Closed Sets — a Formal Definition

Definition (Open Sets)



A subset S of $\mathbb{R}^m$ is said to be **open** if and only if for each $\mathbf{a} \in S$, we can find a real number $\epsilon > 0$ such that all points within distance ϵ of $\mathbf{a}$ are also in S .

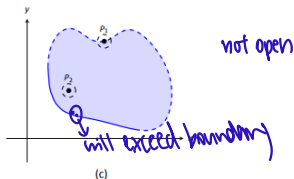
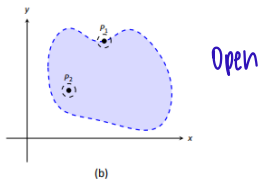
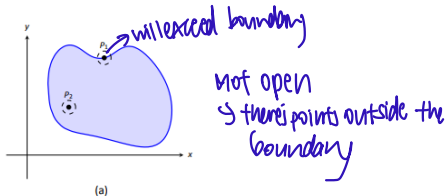
Hence, an open subset of $\mathbb{R}^m$ must contain a sufficiently small ball centered at each of its points.

Definition (Closed Sets)

A subset S of $\mathbb{R}^m$ is said to be **closed** if and only if it contains all of its limit points.

Open and Closed Sets — a Visualization

Question: Which of these three subsets of $\mathbb{R}^2$ are open/closed/neither?



Recall that the derivative of a real-valued function f whose domain is an interval I at a real number $a \in I$ is defined to be the limit

$$f'(a) \stackrel{\text{df}}{=} \lim_{x \rightarrow a} \underbrace{\frac{f(x) - f(a)}{x - a}}_{\text{Difference quotient}}. \quad (\text{If it exists.})$$

To define the partial derivatives of a real-valued function f of several real variables, the main idea is to compute the difference quotient of f with respect to one variable while keeping the other variables fixed.

Partial Derivatives in Two Dimensions — a Formal Definition

Definition (Partial Derivatives in Two Dimensions)

Let f be a real-valued function of two real variables with an **open** domain D .

The **x -** and **y -partial derivatives** of f , denoted by f_x and f_y respectively, are the functions defined by

$$\text{Dom}(f_x) = \left\{ (a, b) \in D \mid \lim_{x \rightarrow a} \frac{f(x, b) - f(a, b)}{x - a} \text{ exists} \right\},$$
$$f_x(a, b) \stackrel{\text{df}}{=} \lim_{x \rightarrow a} \frac{f(x, b) - f(a, b)}{x - a} = \frac{d}{dx}[f(x, b)]|_{x=a}; \text{ fix } y\text{-coordinate}$$

$$\text{Dom}(f_y) = \left\{ (a, b) \in D \mid \lim_{y \rightarrow b} \frac{f(a, y) - f(a, b)}{y - b} \text{ exists} \right\},$$
$$f_y(a, b) \stackrel{\text{df}}{=} \lim_{y \rightarrow b} \frac{f(a, y) - f(a, b)}{y - b} = \frac{d}{dy}[f(a, y)]|_{y=b}. \text{ fix } x\text{-coordinate}$$

Alternative notation: $f_x = \frac{\partial f}{\partial x}$ and $f_y = \frac{\partial f}{\partial y}$. *for more than 1 variable involved.*

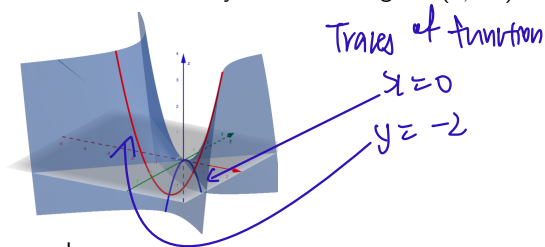
Example

Define $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $f(x, y) \stackrel{\text{df}}{=} x^2 - y^2$.

- (a) Find the rate of change of f at the point $(0, -2)$ in the positive x -direction.
- (b) Find the rate of change of f at the point $(0, -2)$ in the positive y -direction.

Solution

- Let us first use GeoGebra to visualize the x - and y -rates of change at $(0, -2)$:



- $f_x(0, -2) = \frac{d}{dx}[f(x, -2)]|_{x=0} = \frac{d}{dx}[x^2 - 4]|_{x=0} = 2x|_{x=0} = 0.$
- $f_y(0, -2) = \frac{d}{dy}[f(0, y)]|_{y=-2} = \frac{d}{dy}[-y^2]|_{y=-2} = -2y|_{y=-2} = 4.$

Example

Define $f : \mathbb{R} \times (\mathbb{R} \setminus \{0\}) \rightarrow \mathbb{R}$ by $f(x, y) \stackrel{\text{df}}{=} e^{xy} + \frac{x}{y}$.

Problem: Determine f_x and f_y .

$$\frac{\partial}{\partial x} [e^{xy}] = e^{xy} \cdot \frac{\partial}{\partial x} (xy) = e^{xy} \cdot y$$

Solution

- f_x is defined on $\mathbb{R} \times (\mathbb{R} \setminus \{0\})$ and given by $f_x(x, y) = ye^{xy} + \frac{1}{y}$.
- f_y is defined on $\mathbb{R} \times (\mathbb{R} \setminus \{0\})$ and given by $f_y(x, y) = xe^{xy} - \frac{x}{y^2}$.

Example

Define $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f(x, y) \stackrel{\text{df}}{=} \begin{cases} \frac{x^3}{x^2 + y^2}, & \text{if } (x, y) \neq (0, 0); \\ 0, & \text{if } (x, y) = (0, 0). \end{cases}$$

$$\frac{d}{dx}[f(x, 0)]|_{x=0}$$

$$\frac{d}{dy}[f(0, y)]|_{y=0}$$

?

Problem: Evaluate $f_x(0, 0)$ and $f_y(0, 0)$.

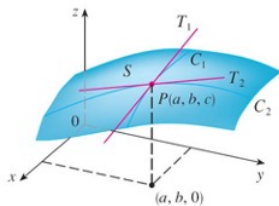
Solution

- For all $x \in \mathbb{R} \setminus \{0\}$, we have $f(x, 0) = \frac{x^3}{x^2 + 0^2} = \frac{x^3}{x^2 + 0} = \frac{x^3}{x^2} = x$.
- As $f(0, 0) = 0$, we find, for all $x \in \mathbb{R}$, that $f(x, 0) = x$.
- Therefore, $f_x(0, 0) = \frac{d}{dx}[f(x, 0)]|_{x=0} = \frac{d}{dx}[x]|_{x=0} = 1|_{x=0} = 1$.
- For all $y \in \mathbb{R} \setminus \{0\}$, we have $f(0, y) = \frac{0^3}{0^2 + y^2} = \frac{0}{0 + y^2} = \frac{0}{y^2} = 0$.
- As $f(0, 0) = 0$, we find, for all $y \in \mathbb{R}$, that $f(0, y) = 0$.
- Therefore, $f_y(0, 0) = \frac{d}{dy}[f(0, y)]|_{y=0} = \frac{d}{dy}[0]|_{y=0} = 0|_{y=0} = 0$.

Partial Derivatives in Two Dimensions — a Geometric Interpretation

Let f be a real-valued function of two real variables with an open domain D .

Suppose that the partial derivatives of f exist at $(a, b) \in D$.



The line T_1 is tangent to the curve C_1 given by $z = f(x, b)$ at the point $(a, b, f(a, b))$.

The slope of T_1 is $f_x(a, b)$. *→ x-derivative at (a,b)*

The line T_2 is tangent to the curve C_2 given by $z = f(a, y)$ at the point $(a, b, f(a, b))$.

The slope of T_2 is $f_y(a, b)$. *→ y-derivative at (a,b)*

two lines determine a plane! ⇒ best way to approximate linearly at (a,b)

T_1 and T_2 span what is known as the **tangent plane** to the graph surface of f at (a, b) .